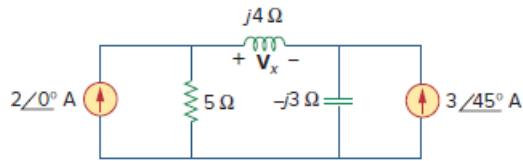


Problem

Use nodal analysis to find V_x in the circuit shown in Fig. 10.65.

Figure 10.65:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

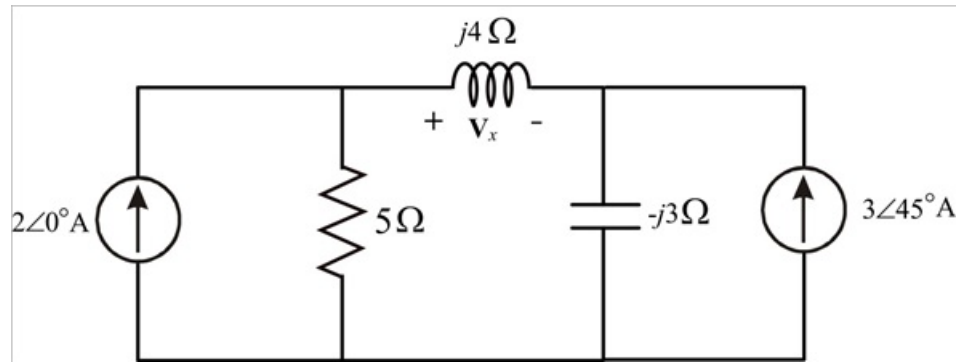


Figure 1

Step 2 of 5

The modified circuit diagram of Figure 1 is shown in Figure 2:

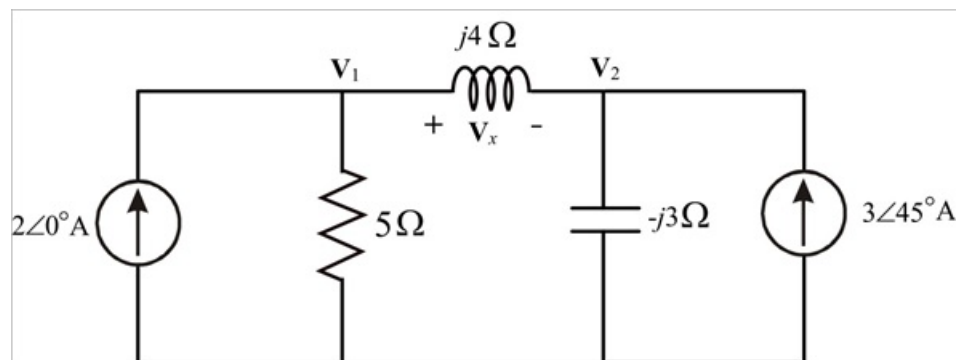


Figure 2

Step 3 of 5

Apply Kirchhoff's current law at node V_1 .

$$\frac{V_1}{5\Omega} - 2\angle 0^\circ + \frac{V_1 - V_2}{j4\Omega} = 0$$

$$\frac{V_1}{5\Omega} - 2\angle 0^\circ + \frac{V_1}{j4\Omega} - \frac{V_2}{j4\Omega} = 0$$

$$V_1 \left(\frac{1}{5\Omega} + \frac{1}{j4\Omega} \right) - \frac{V_2}{j4\Omega} = 2\angle 0^\circ$$

$$V_1 (0.2 - j0.25) + j0.25V_2 = 2\angle 0^\circ \dots\dots (1)$$

Apply Kirchhoff's current law at node V_2 .

$$\frac{V_2}{-j3\Omega} - 3\angle 45^\circ + \frac{V_2 - V_1}{j4\Omega} = 0$$

$$V_2 \left(\frac{1}{-j3\Omega} + \frac{1}{j4\Omega} \right) - \frac{V_1}{j4\Omega} = 3\angle 45^\circ$$

$$V_2 (j0.0833) + j0.25V_1 = 3\angle 45^\circ$$

$$j0.25V_1 + V_2 (j0.0833) = 3\angle 45^\circ \dots\dots (1)$$

Step 4 of 5

Use crammer's rule to solve equation (1) and equation (2).

$$\begin{bmatrix} 0.2 - j0.25 & j0.25 \\ j0.25 & j0.0833 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2\angle 0^\circ \\ 3\angle 45^\circ \end{bmatrix}$$

$$\begin{bmatrix} 0.2 - j0.25 & j0.25 \\ j0.25 & j0.0833 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2.12 + j2.12 \end{bmatrix}$$

Calculate the value of V_1 :

$$V_1 = \frac{\Delta_1}{\Delta}$$

$$= \frac{\begin{vmatrix} 2 & j0.25 \\ 2.12 + j2.12 & j0.0833 \end{vmatrix}}{\begin{vmatrix} 0.2 - j0.25 & j0.25 \\ j0.25 & j0.0833 \end{vmatrix}}$$

$$= \frac{2(j0.0833) - (2.12 + j2.12)(j0.25)}{(0.2 - j0.25)(j0.0833) - (j0.25)(j0.25)}$$

$$= \frac{0.53 - j0.3634}{0.0833 + j0.0166}$$

$$= 5.277 - j5.416$$

Step 5 of 5

Calculate the value of $\mathbf{V}_1$:

$$\begin{aligned}\mathbf{V}_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{\begin{vmatrix} 0.2 - j0.25 & 2 \\ j0.25 & 2.12 + j2.12 \end{vmatrix}}{\begin{vmatrix} 0.2 - j0.25 & j0.25 \\ j0.25 & j0.0833 \end{vmatrix}} \\ &= \frac{(2.12 + j2.12)(0.2 - j0.25) - 2(j0.25)}{(0.2 - j0.25)(j0.0833) - (j0.25)(j0.25)} \\ &= \frac{0.254 - j0.606}{0.0833 + j0.0166} \\ &= 9.62 - j9.192\end{aligned}$$

Calculate the value of $\mathbf{V}_x$:

$$\begin{aligned}\mathbf{V}_x &= \mathbf{V}_1 - \mathbf{V}_2 \\ &= (5.277 - j5.416) - (9.62 - j9.192) \\ &= (5.277 - j5.416) - (9.62 - j9.192) \\ &= -4.343 + j3.776 \\ &= 5.754 \angle 138.9^\circ\end{aligned}$$

Therefore, the value of $\mathbf{V}_x$ is $\boxed{5.754 \angle 138.9^\circ}$.